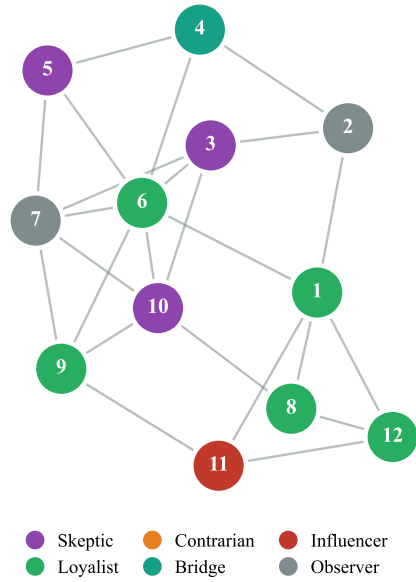


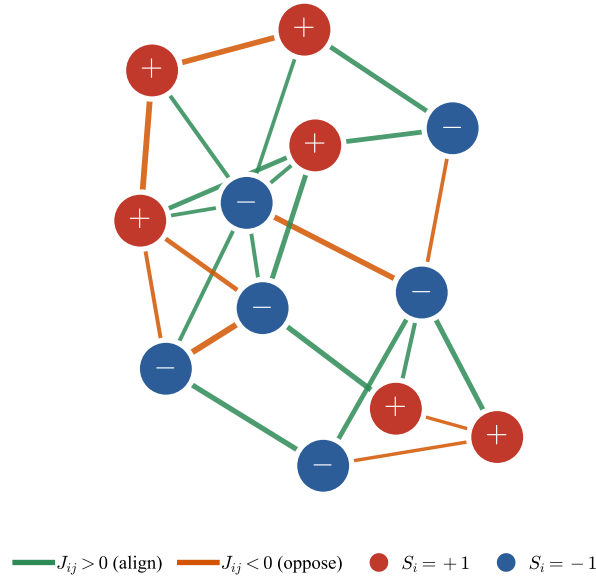
On the Statistical Mechanics of Multi-Agent Interactions

LLM agents with personas on a social graph $\rightarrow$ spin-glass mapping $\rightarrow$ replica & cavity analysis $\rightarrow$ predicted phase diagram of opinion dynamics

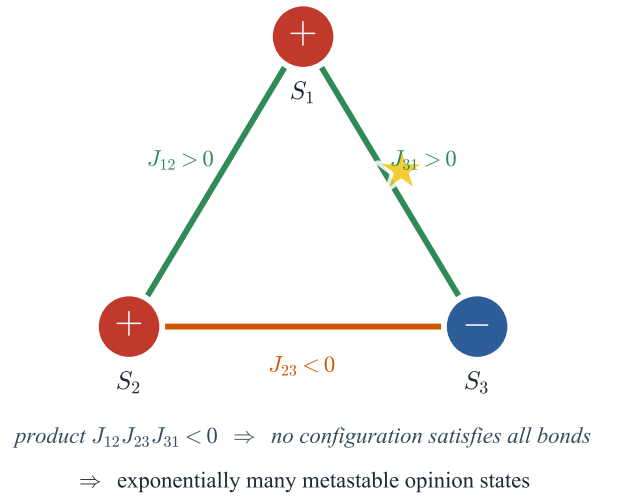
(a) LLM agents with personas form graph



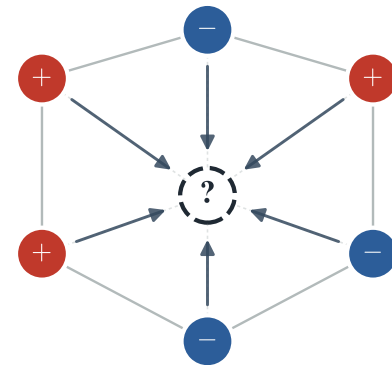
(b) $\Rightarrow$ Spin-glass Hamiltonian $H_J(S) = -\sum J_{ij}S_iS_j - \sum h_iS_i$



(c) Frustration $\Rightarrow$ many local minima

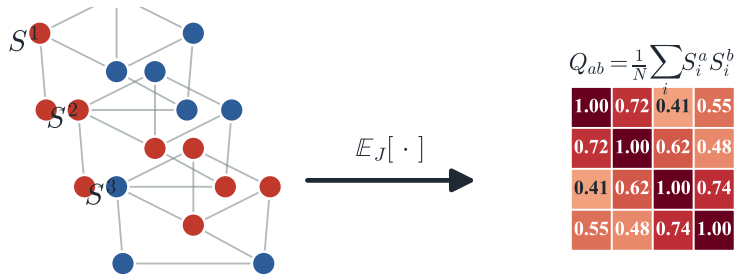


(e) Cavity method: leave-one-out + messages



messages $\eta_{i \rightarrow 0}(S_0) \propto \psi_i(S_i) \prod_{k \in \partial i \setminus 0} \sum_{S_k} \psi_{ik} \eta_{k \rightarrow i}$
 $\Rightarrow$ Belief Propagation (sparse) / AMP (dense)

(d) Replica method: n copies share one J



(f) Predicted phase diagram & overlap distributions

